

Method hierarchy

Capacity-Balance MCS

Baseline

Classical hourly capacity check · no storage

Naive Storage MCS

High EUE bias

Proactive discharge; depletes before scarcity

SOC-Floor Storage MCS

Residual EUE bias

Reserve floor limits proactive discharge

Emergency-Only Storage MCS

EUE-aligned

Reliability dispatch; discharge only at shortfall

Event-Window Storage MCS

EUE-aligned

Windowed LP around screened scarcity windows

Full-Year ED

Continuous-dispatch comparison

Full-year economic dispatch LP · continuous dispatch

PCM-UCED

Operational benchmark

Full-year unit commitment MILP · operational benchmark

Increasing computational cost